「Def. Let V be a real or Complex vector space. A function $f: V\times V \to F$ such that for all $\alpha, \beta, \gamma \in V$ and $c \in F$, $\begin{cases} f(c\cdot\alpha+\beta, \gamma) = c\cdot f(\alpha,\gamma) + f(\beta,\gamma) \\ f(\alpha, c\cdot\beta+\gamma) = \bar{c}\cdot f(\alpha,\beta) + f(\alpha,\gamma) \end{cases}$
is called the Sesqui-linear form.

「Thm. Let V be a finite-dimensional inner product space, and f be a sesqui-linear form on V. Then, there is a unique linear operator T on such that

$$f(\alpha,\beta) = \langle T(\alpha), \beta\rangle \text{ for all } \alpha,\beta\in V.$$

Moreover, a map $f \mapsto T$ is an Isomorphism.

proof. Given $\beta \in V$, $\alpha \mapsto f(\alpha,\beta)$ is a linear functional on V.
By the Riesz Representation Theorem, there exists a unique vector $\beta' \in V$ such that

$$f(\alpha,\beta) = \langle \alpha, \beta'\rangle \text{ for all } \alpha\in V.$$

Now, let $U: V\to V$ be an mapping β to β'. Then, given $\alpha,\beta,\gamma\in V$ and $c\in F$,

$f(\alpha, c\beta+\gamma) = \langle \alpha, U(c\beta+\gamma)\rangle$ and

$$= \bar{c}\cdot f(\alpha,\beta) + f(\alpha,\gamma) = \bar{c}\langle\alpha, U(\beta)\rangle + \langle \alpha, U(\gamma)\rangle = \langle \alpha, c\cdot U(\beta) + U(\gamma)\rangle$$

Since α was arbitrary, U is linear. Since V is finite, adjoint of U exists.
put $T = U^*$. Then, the T is we desired. The uniqueness is clear.
In sum, for each form f, there is a unique linear operator T_f such that

$$f(\alpha,\beta) = \langle T_f(\alpha), \beta\rangle$$ ┘ Existence and Uniqueness.

Given forms f and g and scalar c,

$(cf+g)(\alpha,\beta) = \langle T_{cf+g}(\alpha), \beta\rangle$ and

$$= cf(\alpha,\beta) + g(\alpha,\beta) = \langle (cT_f + T_g)(\alpha), \beta\rangle$$

So $T_{cf+g} = cT_f + T_g$ ┘ $f\mapsto T_f$ is linear
And surjective is clear, and $T_f = 0 \iff f = 0$, so injective. ▨

Corollary. Given forms $f, g : V \times V \to F$, define

$$\langle f, g \rangle \overset{\text{def}}{=} Tr(T_f T_g^*) \overset{(*)}{=} \sum_{j,k} f(\alpha_k, \alpha_j) \cdot \overline{g(\alpha_k, \alpha_j)}$$

is an inner product on $V \times V$ such that $(*)$, for any orthonormal basis $\{\alpha_1, \cdots, \alpha_n\}$ for V.

Proof. Given orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$, let $A = [T_f]_\beta$ and $B = [T_g]_\beta$.

Then, by the orthonormality and the above Thm,

$$A_{jk} = \langle T_f(\alpha_k), \alpha_j \rangle = f(\alpha_k, \alpha_j) \text{ and } B_{jk} = \langle T_g(\alpha_k), \alpha_j \rangle = g(\alpha_k, \alpha_j).$$

Moreover, $[T_f \circ T_g^*]_\beta = [T_f]_\beta \cdot [T_g^*]_\beta = AB^*$, So $Tr(AB^*) = \sum_{j,k} A_{jk} \cdot \overline{B_{jk}}$ gives $(*)$.

Def. Let V be a fin. dim inner product space, with basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$.

For a given form $f : V \times V \to F$, the matrix with entries

$$A_{jk} = f(\alpha_k, \alpha_j)$$

is called the matrix of f in the ordered basis β.

Observe that: for a form $f : V \times V \to F$ and orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$..

Then, $f\left(\sum_s x_s \cdot \alpha_s, \sum_r y_r \alpha_r\right) = \sum_{r,s} \overline{y_r} A_{rs} x_s$, or

$$f(\alpha, \beta) = Y^* A X \quad \text{where } X = [\alpha]_\beta, \ Y = [\beta]_\beta.$$

Thm. Let V be a fin. dim Complex inn. Prod space.

For any form on $V \times V$, there is an orthonormal basis for V,

in which the matrix of f is upper-triangular.

Proof. Since V is fin. dim. Complex. inn. Prod, space, for linear operator $T : V \to V$ s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle,$$

it has an orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ for V s.t. $[T]_\beta$ is upper-triangular. So,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = 0 \quad \text{if } j > k.$$

Def. A form f on V is called Hermitian if
$$f(\alpha,\beta)=\overline{f(\beta,\alpha)} \text{ for all } \alpha,\beta\in V.$$

Note: if there is a linear operator T on V such that $f(\alpha,\beta)=\langle T(\alpha),\beta\rangle$, then $\overline{f(\beta,\alpha)}=\langle\alpha,T(\beta)\rangle=\langle T^*(\alpha),\beta\rangle$, so,

$$f \text{ is Hermitian if and only if } T \text{ is Hermitian.}$$

If T is Hermitian, then $\langle T(\alpha),\alpha\rangle$ is real for all $\alpha\in V$, so $f(\alpha,\alpha)$ is real.

Lemma. Let V be a complex inner product space, and f is a form on V such that
$$f(\alpha,\alpha) \text{ is real, for all } \alpha\in V.$$

Then, f is Hermitian.

Proof. Given $\alpha,\beta\in V$,
$$f(\alpha+\beta,\alpha+\beta)=f(\alpha,\alpha)+f(\beta,\beta)+f(\alpha,\beta)+f(\beta,\alpha)$$
$$f(\alpha+i\beta,\alpha+i\beta)=f(\alpha,\alpha)+f(i\beta,i\beta)-i\cdot f(\alpha,\beta)+if(\beta,\alpha)$$
By the assumption, $f(\alpha,\beta)+f(\beta,\alpha)$ and $-i\cdot f(\alpha,\beta)+i\cdot f(\beta,\alpha)$ are real. Thus,
$$f(\alpha,\beta)+f(\beta,\alpha)=\overline{f(\alpha,\beta)}+\overline{f(\beta,\alpha)} \text{ and } -i\cdot f(\alpha,\beta)+i\cdot f(\beta,\alpha)=i\cdot\overline{f(\alpha,\beta)}-i\cdot\overline{f(\alpha,\beta)}$$

Multiplying i to the right, then we obtain
$$f(\alpha,\beta)-f(\beta,\alpha)=-\overline{f(\alpha,\beta)}+\overline{f(\alpha,\beta)}$$
So, $2f(\alpha,\beta)=2\cdot\overline{f(\alpha,\beta)}$, or f is Hermitian. QED

Combining the above note with this Lemma, if V is fin.dim, then
$$T \text{ is Hermitian if and only if } \langle T(\alpha),\alpha\rangle \text{ is real.}$$

「Principal Axis Theorem.

「Thm. For every Hermitian form f on a fin. dim inn. prod space V, there exists an orthonormal basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ such that

f is represented by a diagonal matrix with real entries.

Proof. Since V is fin. dim, there exists a linear operator T on V s.t

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle.$$

And, by the lemma above, f is Hermitian iff T is Hermitian, so there is an orthonormal basis for V which is consisted by eigenvector of T. Denote $\beta = \{\alpha_1, \cdots, \alpha_n\}$, and c_i be the eigenvalue corresponding to α_i. Then,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = c_k \cdot \delta_{kj},$$

thus proved. (c_k are real by T is Hermitian). ☒

Note: Under the condition hold,

$$f\left(\sum x_j \alpha_j, \sum y_k \alpha_k\right) = \sum_i c_i x_i \overline{y_i}$$